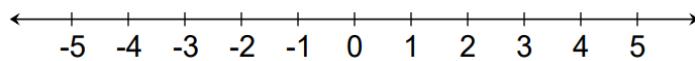


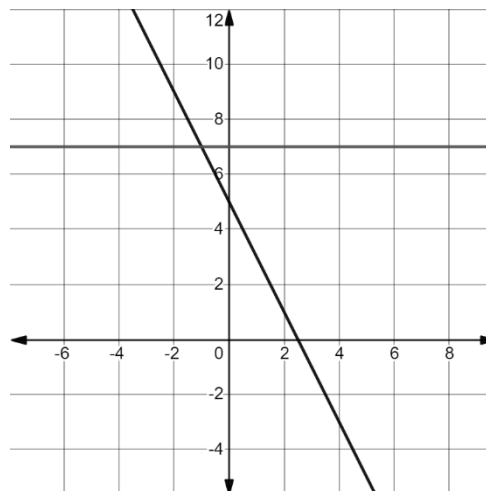
Consider the inequality $-2x + 5 \leq 7$.

1. Solve the inequality.

2. Graph the solution set.



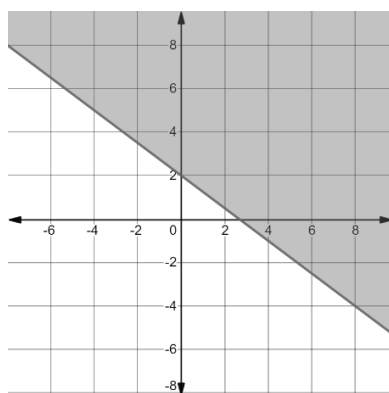
3. The graphs of $y = -2x + 5$ and $y = 7$ are shown on the coordinate plane.



Highlight the part of the graph of $y = -2x + 5$, where the y values are less than or equal to 7.

4. Describe the relationship between the solution to the inequality $-2x + 5 \leq 7$ and the values of x for which the graph of $y = -2x + 5$ is less than or equal to 7.

The graph of the two-variable inequality $y \geq -\frac{3}{4}x + 2$ is shown.



5. The slanted solid line is called the **boundary line** of the inequality. Write the equation of the boundary line in slope-intercept form.
6. How does the equation of the boundary line relate to the inequality?
7. Describe what part(s) of the graph represent the solution set of the inequality $y \geq -\frac{3}{4}x + 2$.
8. Which part of the graph of the solution set of $y \geq -\frac{3}{4}x + 2$ corresponds to each mathematical statement?

Mathematical Statement	Graphical Representation
$y = -3/4x + 2$	☐ boundary line ☐ shaded region
$y > -3/4x + 2$	☐ boundary line ☐ shaded region

Complete the statements.

9. If the points on the line are included in the solution set, the boundary line is
☐ dotted.
☐ solid.
10. If the points on the line are not included in the solution set, the boundary line is
☐ dotted.
☐ solid.